



PRESIDENCY UNIVERSITY

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School of Engineering/School of Computer Science and Engineering

A Project Report on

**NATURAL HUMAN-LIKE ROBOTIC ARM FOR GRIP AND
MANIPULATION**

Submitted in partial fulfillment of the requirements for the course
Design Workshop (ECE 1511)

Submitted by

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Abstract :

This project presents the design and development of a natural human-like robotic arm controlled using a glove embedded with five flex sensors. The system is designed to mimic human hand movement in real time. When the user bends their fingers, the flex sensors detect the motion and send signals to the Arduino microcontroller. Based on these inputs, five servo motors rotate accordingly to move the robotic fingers.

The robotic arm is capable of gripping, lifting, and manipulating lightweight objects. This system can be used in automation, prosthetic research, medical assistance, and hazardous environment handling. The project demonstrates an affordable and efficient human-motion-controlled robotic system.

Introduction :

Robotic arms are widely used in industries and research fields for performing repetitive, dangerous, and precise tasks. Traditional robotic arms require manual programming or remote control systems.

This project introduces a glove-controlled robotic arm that naturally follows human finger movements. A glove fitted with five flex sensors acts as the input device. As the user moves their fingers, the robotic hand replicates the same motion. This creates a natural and intuitive human-machine interaction system.

Motivation:

- Need for smart automation systems.
- Reduce human effort in risky tasks.
- Improve object handling precision.
- Help disabled persons through prosthetic technology.
- Learn embedded systems and robotics.
- Build a low-cost robotic solution.

Objective of the Project:

The main objectives of this project are:

- Design a human-like robotic arm
- Control robotic fingers using hand glove
- Use flex sensors to capture finger motion
- Use Arduino for processing sensor data
- Use servo motors for accurate finger movement
- Perform gripping and manipulation tasks
- Develop a cost-effective prototype

Hardware, Software and tools used:

Hardware Components

- Arduino UNO
- 5 Flex Sensors
- 5 Servo Motors (SG90/MG995)
- Robotic Hand Structure
- Control Glove
- Jumper Wires
- Breadboard / PCB
- External Power Supply / Battery
- Resistors (10k Ω)
- Connecting Threads

Software Tools

- Arduino IDE
- Embedded C/C++ programming

Block diagram & Description:

System Architecture Overview

- The system consists of a glove fitted with 5 flex sensors to detect finger bending.
- Each flex sensor is attached to one finger of the glove.
- When the user bends a finger, the sensor resistance changes.
- These signals are sent to the Arduino UNO through analog pins A0 to A4.
- Arduino reads and processes the sensor values.
- The controller converts sensor data into servo motor angles.
- PWM output signals are generated from digital pins D2 to D6.
- Five servo motors receive the control signals.
- Each servo motor controls one finger of the robotic hand.
- The robotic fingers move according to glove finger motion.
- A 5V external power supply powers the servos and Arduino.
- Common ground connection is maintained for stable operation.
- The robotic hand performs gripping and releasing actions.
- This provides real-time human hand gesture control.

Working Principle

The glove contains five flex sensors mounted on each finger. Flex sensors change their resistance when bent.

When the user bends a finger:

- Flex sensor resistance changes.
- Arduino reads analog values.
- Values are converted into servo motor angles.
- Corresponding servo motor rotates.
- Robotic finger bends similarly.

Results:



- Successful real-time finger motion replication
- Accurate control of all five servo motors
- Smooth opening and closing of robotic fingers
- Effective gripping of lightweight objects
- Natural glove-based control achieved
- Good response speed and repeatability
- Stable operation after calibration

Challenges faced:

- Flex sensor calibration issues
- Different sensor readings for same finger movement
- Servo motor jitter due to low power supply
- Voltage drop when multiple servos run together
- Mechanical alignment problems in robotic fingers
- Improper string/linkage tension affecting movement
- Complex wiring on glove with 5 flex sensors
- Limited gripping strength for heavy objects
- Sensor noise causing unstable readings
- Delay in synchronized finger movement
- Servo angle tuning required for accurate motion

Conclusion:

- The project successfully developed a glove-controlled natural human-like robotic arm using Arduino, 5 flex sensors, and 5 servo motors.
- The robotic hand accurately copied human finger movements in real time and performed gripping actions.
- The system proved to be a low-cost, simple, and effective solution for robotic hand control.
- It can be used in industrial automation, prosthetic hands, medical assistance, and hazardous work environments.
- With future improvements like wireless control, AI vision, and stronger motors, the project can be made more advanced and efficient.

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